

TOPPER CLASSROOM EXAM SERIES

ICSE GRADE 5 SCIENCE

Master Class Answer Notes & Complete Revision Guide

Grounded in the Scanned Science Chapters:

- Chapter 3: Food and Health
- Chapter 4: Pollination
- Chapter 5: Plant Reproduction
- Chapter 6: Solids, Liquids, and Gases

Included Materials (Solved Comprehensive Study-Kit):

1. **Definitive Chapter Notes & Vocabulary:** Full summary, key points, and definitions.
2. **Middle of the Chapter Questions:** All in-text exercises fully solved.
3. **Comprehensive Section I Exercises:** Multiple Choice, Assertion-Reasoning, Fill in the Blanks, and Examples.
4. **Comprehensive Section II Exercises:** Complete Short & Long Answer Questions.
5. **Diagram & Picture Study:** Analytical diagrams described and fully labeled.
6. **Mid-Term Revision Worksheet Solutions:** Every short, long, and analytical question solved to perfection.

ICSE TOPPER EXAMINATION SPECIFICATION: This study package is structured exactly as an ICSE Grade 5 topper prepares for examinations. Every answer is complete and written out in a point-wise format using exact vocabulary from the textbook. Key examiner tips, diagrammatic guidance, and high-scoring structures are annotated to secure full marks (80/80) in the exam.

TOPPER STUDY COMPILATION STRUCTURE

Table of Contents

This study-kit is divided chapter-wise to provide absolute clarity and systematic review. Each section begins with essential facts, followed by textbook questions and exam-style worksheets. Use the diagram guides and examiner tips to structure your sheet on the final exam.

Chapter	Focus Areas	Key Exam Deliverables
Chapter 3: Food and Health	Nutrients (Carbs, Fats, Proteins, Vitamins, Minerals), Balanced Diet, Junk Food, Lifestyle & Deficiency Diseases, Food Adulteration	Diet Chart Pyramids, Deficiency Tables, Solved Adulteration Case Studies
Chapter 4: Pollination	Anatomy of Flower (4 Whorls), Monosexual vs Bisexual Flowers, Self vs Cross Pollination, Pollination Agents	Flower Diagrams, Labeled Flower Parts Worksheet, Agent Mechanism Analysis
Chapter 5: Plant Reproduction	Fertilisation Process, Seed Structure (Bean Seed), Germination, Dispersal Methods, Vegetative Propagation	Seed Germination Stages, Bean Seed Labeled Anatomy, Spore Reproduction
Chapter 6: Solids, Liquids, & Gases	Molecular Arrangement of Matter, Solutions (Solute, Solvent), Separations, Air Composition & Breezes	Molecular Circle Diagrams, Soluble vs Insoluble Separation, Land & Sea Breeze
Mid-Term Revision Worksheet	Integrated Science Assessment: Combined Questions (Section A & B) and Critical Thinking Section	Full 20-Question Long/Short Solved Answers, Reversibility Critical Analysis

EXAMINER WRITING METHODOLOGY: ICSE examiners award marks based on precise keywords and clean formatting. Avoid writing answers in long, unstructured paragraphs. Always use bullet points, bold key definitions, and support long answers with neat, fully-labeled diagrams.

ICSE GRADE 5 SCIENCE NOTES

Chapter 3: Food and Health

I. IMPORTANT THINGS TO KNOW (KEY POINTS)

- Food is vital for survival as it provides energy for growth, cellular repair, and normal functioning.
- The five main components of food are called **nutrients**: carbohydrates, fats, proteins, vitamins, and minerals.
- **Carbohydrates and Fats** provide energy, **Proteins** are for growth and body repair, and **Vitamins and Minerals** protect the body from diseases.
- A **Balanced Diet** contains all essential nutrients, along with water and roughage, in correct proportions.
- **Junk Food** is high in sugar, salt, and fat but highly deficient in nutrients and fibres; eating it regularly leads to obesity and poor digestion.
- **Lifestyle Diseases** (e.g. obesity, diabetes, hypertension) arise from incorrect food habits and lack of physical exercise. **Deficiency Diseases** (e.g. scurvy, goitre, anaemia, rickets) result from lack of specific nutrients in the diet.
- **Food Adulteration** is the illegal act of adding low-quality, unwanted, or unsafe substances (adulterants) to food items to increase quantity or make profit (e.g. adding water to milk, chalk powder to pulses).

Textbook Summary of Food Components:

Nutrient Group	Primary Function	Key Dietary Sources
Carbohydrates	Provide quick and main energy to the body	Cereals (rice, wheat, corn), potato, sugar, fruits
Fats	Provide highly concentrated energy; keep the body warm	Oil, milk, butter, ghee, cheese, cream, dry fruits
Proteins	Essential for growth, muscle-building, and cellular repair	Eggs, meat, fish, milk, cheese, soya beans, pulses
Vitamins	Needed in small amounts for body functions and disease prevention	Fresh fruits, green leafy vegetables, fish, eggs
Minerals (Iron, Calcium, Iodine, etc.)	Needed in small amounts for keeping bones, teeth, and blood healthy	Vegetables, fresh fruits, milk, eggs, iodized salt
Water & Roughage	Roughage (plant fibre) prevents constipation. Water aids digestion & transport.	Whole grains, salads, raw vegetables, drinking water

II. NEW WORDS DEFINITIONS

- **Nutrient:** An essential chemical component of food that is necessary for growth, maintenance, and repair of the body.
- **Roughage:** The indigestible fibrous portion of plant food that helps in the smooth passage of food and prevents constipation.
- **Balanced Diet:** A diet that provides all five nutrient groups, along with water and roughage, in the correct proportions.
- **Adulterants:** Cheap, unwanted, and unsafe substances added to food items to increase quantity, presentation, or profit.
- **Lifestyle Diseases:** Diseases that are caused due to improper food habits, lack of physical activity, and unhealthy lifestyle.
- **Deficiency Diseases:** Non-communicable diseases caused by a long-term shortage of a specific vitamin or mineral in a person's diet.

III. IN-TEXT MIDDLE OF CHAPTER QUESTIONS

Q1. Match the following (Page 39):

Question:

1. Carbohydrate
2. Vitamins
3. Roughage
4. Protein
5. Junk Food

Answer:

1. Carbohydrate — **(b) Provides energy.**
2. Vitamins — **(d) Prevents diseases.**
3. Roughage — **(e) Undigested portion of food.**
4. Protein — **(a) Growth and repair of body.**
5. Junk Food — **(c) Burgers, pizzas, patties.**

IV. EXERCISES: SECTION I SOLUTIONS

A. Choose the correct option:

- **Question:** Which of these is the main component of food?

a. Fats b. Carbohydrates c. Proteins d. All of these

Topper Answer: d. All of these

- **Question:** Oil, milk, butter, and cheese mainly contain

a. roughage b. fats c. proteins d. vitamins

Topper Answer: b. fats

• **Question:** Which among these is not a lifestyle disease?

a. Diabetes b. Anaemia c. Scurvy d. Obesity

Topper Answer: c. Scurvy (It is caused by lack of Vitamin C, a deficiency disease)

• **Question:** Rickets is caused due to the lack of

a. vitamin A b. vitamin B c. vitamin C d. vitamin D

Topper Answer: d. vitamin D

• **Question:** Goitre occurs due to the lack of

a. calcium b. iron c. iodine d. protein

Topper Answer: c. iodine

• **Question:** Which among these is not an adulterant?

a. Milk b. Chalk powder c. Marble chips d. None of these

Topper Answer: a. Milk (Adulterants are unwanted things added; milk itself is the pure food item)

B. Assertion and Reasoning Questions:

• **Question: Assertion (A):** Food adulteration is done to improve the nutritional value of food.

Reason (R): Food adulteration involves adding unwanted and unsafe substances.

Topper Answer: d. A is False and R is True.

Topper Explanation: Assertion (A) is completely false because food adulteration decreases and destroys the nutritional value of food. It is done illegally to increase food weight/volume to cheat buyers and make profits.

Reason (R) is true as adulteration utilizes unsafe and unwanted substances like chalk powder or water.

• **Question: Assertion (A):** Eating a lot of fruits and vegetables can prevent deficiency diseases.

Reason (R): Fruits and vegetables are rich in vitamins and minerals.

Topper Answer: a. Both A and R are True and R is the correct explanation of A.

Topper Explanation: Deficiency diseases are caused by lack of vitamins/minerals. Fruits and vegetables are 'protective foods' rich in these nutrients, so eating them prevents deficiency diseases.

C. Choose the correct option to fill in the blank:

• **Question:** There are (four/five) main components of food. — **Answer: five**

• **Question:** Minerals are needed in (large/small) amounts by our body. — **Answer: small**

• **Question:** Roughage includes mostly (water/fibres). — **Answer: fibres**

• **Question:** Junk food contains high amount of (sugar/roughage). — **Answer: sugar**

• **Question:** It is (important/not important) to eat a balanced diet. — **Answer: important**

D. Give two examples of the following:

• **1. Vitamin-rich foods:** Citrus fruits (oranges, lemons), green leafy vegetables (spinach)

• **2. Carbohydrate-rich foods:** Cereals (rice, wheat), potatoes

- **3. Junk foods:** Burgers, pizzas, potato chips
- **4. Fermented foods:** Yogurt, idli/dosa, dhokla, panta bhaat

V. EXERCISES: SECTION II SOLUTIONS

E. Short Answer Questions (Page 48):

Q.1) Why should roughage be present in our diet?

Answer: Roughage consists of dietary plant fibres. It is crucial because it helps in the smooth passage of food through the digestive tract, adds bulk to stools, prevents constipation, and aids in healthy bowel movements.

Q.2) How does a balanced diet help in the growth and development of the mind?

Answer: A balanced diet provides essential vitamins, iron, and minerals that nourish the nervous system. It enhances brain performance, boosts concentration and memory power, and keeps us mentally active for daily tasks.

Q.3) Write any three harmful effects of junk food.

Answer: 1) **Obesity:** High sugar and fats lead to abnormal weight gain.
2) **Deficiency of Nutrients:** It lacks vitamins, minerals, and fibres, causing nutritional deficiencies.
3) **Digestion and Disease Issues:** High salt and sugar cause poor digestion and increase the risk of blood pressure and diabetes.

Q.4) How is anaemia caused?

Answer: Anaemia is caused by a deficiency of red blood cells or haemoglobin in the blood, which results from a lack of **iron** in our diet, general malnutrition, excessive blood loss, or worm infestation in the stomach.

Q.5) What are the symptoms of night blindness?

Answer: The primary symptom of night blindness (caused by lack of Vitamin A) is the inability to see clearly in dim light, at sunset, or during the night.

F. Long Answer Questions (Page 48):

Q.1) You're packing a lunch for a long hike. Considering the importance of a balanced diet, what kind of foods would you include to ensure you have enough energy for the entire hike?

Answer: For a long hike, I would select a combination of foods that provide both instant and long-lasting energy, while keeping the meal easy to carry:

- 1) **Carbohydrates (For Main Energy):** Whole wheat chapatis or wraps, and bananas. These provide slow-release and instant energy to sustain physical activity.
- 2) **Proteins (For Muscle Recovery):** Roasted chickpeas, paneer/cheese cubes, or boiled eggs to help rebuild muscles after strenuous climbing.
- 3) **Healthy Fats (Concentrated Energy):** A mix of almonds, walnuts, and raisins. Fats provide highly concentrated energy, and dry fruits don't spoil easily.
- 4) **Hydration:** Adequate drinking water and a slice of lemon/orange for electrolytes, preventing dehydration and fatigue.

Q.2) You're helping your family switch to a healthier lifestyle. List three specific ways you could modify your family's diet to make it healthier.

Answer: To improve my family's dietary health, I would suggest the following three modifications:

- 1) **Reduce Sugar, Salt, and Oil:** Use minimum amount of oil, avoid deep frying (prefer steaming or baking), and restrict the addition of artificial sugars in beverages.
- 2) **Eat Fresh and Unprocessed Foods:** Introduce fresh seasonal fruits (like apples and guavas) daily. Substitute polished white cereals with fiber-rich whole grains.
- 3) **Incorporate Sprouts and Fermented Foods:** Add sprouts of gram/moong dal to meals, and consume fermented foods like yogurt, idli, or dhokla regularly, which contain gut-friendly bacteria that aid digestion.

Q.3) Differentiate between lifestyle diseases and deficiency diseases.

Answer: A topper structures this difference as a clear comparative table:

Feature	Lifestyle Diseases	Deficiency Diseases
Primary Cause	Caused by incorrect food habits, lack of physical activity, and poor lifestyle choice.	Caused by the long-term shortage/lack of a specific vitamin or mineral in the daily diet.
Nature	Non-communicable; lifestyle-dependent.	Non-communicable; nutrient-dependent.
Cure	Corrected by regular exercise, healthy diet, proper sleep, and reducing junk food.	Cured by consuming foods rich in the missing nutrient or taking supplements.
Examples	Obesity, Type-2 Diabetes, Hypertension (High Blood Pressure), Heart Diseases.	Scurvy (Vitamin C), Goitre (Iodine), Rickets (Vitamin D), Anaemia (Iron).

Q.4) Write the symptoms, causes, and prevention of any two deficiency diseases.

Answer: I will list three major deficiency diseases in a high-scoring table format:

Deficiency Disease	Nutrient Deficient	Common Symptoms	Prevention Methods
Scurvy	Vitamin C	Swollen gums that bleed easily; slow healing of wounds.	Regularly eat citrus fruits like amla, oranges, lemons, and fresh tomatoes.
Goitre	Iodine	Swelling of the thyroid gland in the neck; mental slowdown.	Use iodized salt in cooking and consume iodine-rich foods.
Rickets	Vitamin D & Calcium	Softening and weakening of bones; bowlegs; knock-knees.	Consume milk, cheese, fish, eggs, and spend time in morning sunlight.

Q.5) Give common examples of food adulteration.

Answer: Adulteration involves illegally adding low-grade materials to food. Common examples are:

- 1) **Milk:** Addition of water to increase volume, or starch and urea to maintain thickness.
- 2) **Cereals and Pulses:** Mixing with chalk powder, marble chips, and stone chips to increase weight.
- 3) **Spices:** Adulterating turmeric powder and chilli powder with lead pigments or brick powder to enhance color artificially.

VI. PICTURE STUDY SOLUTIONS (Page 49)

Picture Study: Food Pyramid

1. Food Item and Nutrient Labeling of the Pyramid:

- **Top Level (Smallest):** Oils, butter, ghee, cream, sweets. *Nutrient:* **Fats** (provides energy).
- **Second Level:** Milk, cheese, eggs, fish, meat, pulses, beans. *Nutrient:* **Proteins** (growth and repair).
- **Third Level:** Fresh fruits, leafy green vegetables. *Nutrient:* **Vitamins & Minerals** (disease protection).
- **Bottom Level (Largest):** Bread, rice, cereals, potatoes, chapatis. *Nutrient:* **Carbohydrates** (main quick energy).

2. What does the given picture depict?

Answer: The picture depicts a **Balanced Diet Food Pyramid**, which serves as a visual guide to showing the types and proportions of food groups we must consume daily to maintain optimum health.

3. What is the importance of eating such a diet that contains all the nutrients in adequate amount?

Answer: Eating a diet with adequate nutrients is crucial because:

- " 1) It ensures proper growth and physical/mental development of children.
- " 2) It maintains healthy body mass, providing energy without causing obesity.
- " 3) It boosts the immune system, preventing deficiency diseases like scurvy or night blindness.

TOPPER'S EXAM TIP (CHAPTER 3): In ICSE Grade 5 exams, the 'Case Study' about spices adulteration with sawdust or brick powder often appears. Always highlight that eating these causes severe digestion issues and organ damage. Mention the Food Safety and Standards Authority of India (FSSAI) as the governing body keeping checks on food quality.

ICSE GRADE 5 SCIENCE NOTES

Chapter 4: Pollination

I. IMPORTANT THINGS TO KNOW (KEY POINTS)

- A flower is the **reproductive part** of a plant that is responsible for producing fruits and seeds.
- A typical flower consists of four concentric rings called whorls: **calyx (sepals)**, **corolla (petals)**, **androecium (stamens)**, and **gynoecium (pistil/carpel)**.
- The **Calyx** (sepals) protects the flower in the bud stage. The **Corolla** (bright petals) attracts insects for pollination.
- The **Androecium** is the male reproductive unit. Each stamen has a **filament** and an **anther** that produces pollen grains containing male gametes.
- The **Gynoecium/Pistil** is the female reproductive unit. It has three parts: **stigma** (sticky top), **style** (tube), and **ovary** (swollen base containing ovules).
- **Bisexual flowers** contain both stamen and pistil (e.g. rose, hibiscus). **Unisexual flowers** contain only stamens or only pistils (e.g. papaya, watermelon).
- **Pollination** is the transfer of pollen grains from the anther to the stigma by agents like wind, water, insects, and birds.
- **Self-pollination** occurs in the same flower or on different flowers of the same plant. **Cross-pollination** occurs between different flowers on different plants of the same kind.

II. NEW WORDS DEFINITIONS

- **Pedice**l: The green stalk that attaches the flower to the main stem.
- **Thalamus**: The swollen tip of the pedicel upon which all the whorls of the flower rest.
- **Calyx**: The outermost whorl, composed of green leaf-like sepals.
- **Corolla**: The second whorl, made of colorful, scented leaf-like petals.
- **Androecium**: The male reproductive whorl of the flower, consisting of stamens.
- **Gynoecium (Pistil)**: The female reproductive whorl, consisting of carpels (pistils).
- **Pollination**: The physical transfer of pollen grains from the male anther to the female stigma.

III. IN-TEXT MIDDLE OF CHAPTER QUESTIONS (Page 56)

Q1. Write T for True and F for False:

- **Question:** The outermost whorl of a flower is called the corolla. — **Topper Answer: False** (It is called the calyx)

- **Question:** The androecium is called the male reproductive part of a plant. — **Topper Answer: True**
- **Question:** A pistil has three parts—stigma, style, and ovary. — **Topper Answer: True**
- **Question:** Tulip is an example of a monosexual flower. — **Topper Answer: False** (Tulip has both stamens and pistil, so it is a bisexual flower)

IV. EXERCISES: SECTION I SOLUTIONS

A. Choose the correct option:

- **Question:** Which of the following is not a part of a flower?

a. Corolla b. Calyx c. Stamen d. Petiole

Topper Answer: d. Petiole (A petiole is the stalk of a leaf; a pedicel is the stalk of a flower)

- **Question:** Which of the following is present inside the anther of a stamen?

a. Pollen grains b. Male reproductive cells c. Male gametes d. All are correct

Topper Answer: d. All are correct

- **Question:** What does the stigma of the flower receive during pollination?

a. Style b. Pollen grains c. Ovules d. Seeds

Topper Answer: b. Pollen grains

- **Question:** Which of the following is not an example of a monosexual flower?

a. Hibiscus flower b. Papaya flower c. Pumpkin flower d. Watermelon flower

Topper Answer: a. Hibiscus flower (Hibiscus is a classic bisexual flower)

- **Question:** Pollination leads to the formation of

a. Ovules b. Seeds c. Male gametes d. Pollen grains

Topper Answer: b. Seeds (Following pollination, fertilisation takes place, forming seeds)

B. Assertion and Reasoning Questions:

- **Question: Assertion (A):** The stamen produces pollen grains containing male gametes.

Reason (R): The stamen is the male reproductive unit of a flower.

Topper Answer: a. Both A and R are True and R is the correct explanation of A.

Topper Explanation: The stamen is the male organ (androecium) of a flower, which is why its primary function is to manufacture pollen grains that carry the male reproductive cells.

- **Question: Assertion (A):** Petals are brightly coloured in flowers to attract insects.

Reason (R): Brightly coloured petals serve to attract insects that aid in pollination.

Topper Answer: a. Both A and R are True and R is the correct explanation of A.

Topper Explanation: Insects are visual pollinators. Colorful petals act as a visual advertisement to guide them to the flower's nectar, which enables pollen to stick to their bodies, aiding in pollination.

C. Name the following (Page 59):

- **Question:** 1. The colourful and reproductive part of a plant — **Answer: Flower**
- **Question:** 2. The various parts of a flower — **Answer: Calyx (sepals), Corolla (petals), Androecium (stamens), Gynoecium (pistil)**
- **Question:** 3. The whorl that protects the flower in its bud stage — **Answer: Calyx (sepals)**
- **Question:** 4. The long and slender part of the stamen — **Answer: Filament**
- **Question:** 5. The female reproductive unit of the gynoecium — **Answer: Pistil / Carpel**

V. EXERCISES: SECTION II SOLUTIONS**D. Short Answer Questions (Page 59):**

Q.1) What are the two reproductive parts of a flower?

Answer: The male reproductive part is the **stamen** (Androecium) and the female reproductive part is the **pistil/carpel** (Gynoecium).

Q.2) What is the importance of corolla present in a flower?

Answer: The corolla (ring of colorful petals) is crucial because its bright colors and scent attract insects like bees and butterflies, which act as primary agents for pollination. It also protects the inner reproductive parts.

Q.3) Mention the various parts of the androecium.

Answer: The androecium is made of stamens. Each stamen has two parts: a long slender stalk called a **filament**, and a terminal double-lobed sac called an **anther** (which contains pollen grains).

Q.4) What is a bisexual flower?

Answer: A flower that contains both male reproductive parts (stamens) and female reproductive parts (pistil) within the same flower structure. Examples: Rose, Hibiscus, Lily, and Mustard.

Q.5) Define unisexual flowers with examples.

Answer: Flowers that contain either only the male part (stamens) or only the female part (pistil), but not both, are called unisexual or monosexual flowers. Examples: Papaya, Pumpkin, Cucumber, and Watermelon.

E. Long Answer Questions (Page 59):

Q.1) Describe the different parts of a flower.

Answer: A complete flower stands on a green stalk called the **pedicel**, which swells at the top into the **thalamus**. The four concentric whorls arranged from outside to inside are:

- 1) **Calyx (Sepals):** The outermost ring. It consists of green, leaf-like sepals that shield and protect the flower during its delicate bud stage. Sepals also carry out photosynthesis.
- 2) **Corolla (Petals):** The second whorl. It consists of large, brightly colored petals. Petals contain scent-producing glands and nectar to attract insects for pollination.
- 3) **Androecium (Stamens):** The male whorl. Composed of stamens. Each stamen has a slender stalk (filament) supporting a pollen-producing head (anther). Pollen grains contain the male reproductive cells.
- 4) **Gynoecium / Pistil (Carpel):** The innermost female whorl. Each carpel has a sticky landing pad at the top

(**stigma**) to catch pollen, a hollow connecting tube (**style**), and a swollen base (**ovary**) containing tiny bead-like structures (**ovules**) which house the female eggs.

Q.2) Define pollination. Also, mention the two types of pollination.

Answer: Pollination is defined as the physical process of transferring pollen grains from the pollen-producing anther of a stamen to the receptive stigma of a pistil. It is the necessary precursor to fertilization. There are two types of pollination:

- 1) **Self-Pollination**: This occurs when pollen grains from the anther fall directly on the stigma of the same flower, or on the stigma of another flower on the same plant. Example: Pea plant.
- 2) **Cross-Pollination**: This occurs when pollen grains from the anther of one flower are carried to the stigma of a flower on a different plant of the same species. This requires insect, wind, or water vectors. Example: Hibiscus, Rose, Papaya.

VI. PICTURE STUDY & WORKSHEET SOLUTIONS (Page 60-62)

Picture Study: Flower Anatomy

1. Labeling of the numbered parts:

- (i) Pedicel (stalk)
- (ii) Sepal (calyx)
- (iii) Style (gynoecium tube)
- (iv) Anther (pollen sac)
- (v) Filament (stamen stalk)
- (vi) Stigma (sticky top)
- (vii) Petal (corolla)
- (viii) Ovary (swollen base)

2. In which part are ovules present?

Answer: Ovules are present inside the **Ovary (viii)**.

3. What is the function of part (iii) (Style)?

Answer: The style acts as a mechanical passageway. It elevates the sticky stigma (vi) to receive pollen and provides the channel through which the germinating pollen tube grows down to reach the ovary (viii).

4. What does part (iv) (Anther) produce?

Answer: The anther produces powdery **pollen grains**, which contain the male gametes (gametic cells).

5. Name the process that is involved in the transfer of pollen from part (iv) to part (vi).

Answer: This process is called **Pollination**.

Worksheet: Flower Parts Functions (Page 62)

Part	Main Function of the Part
1. Sepal (Calyx)	Protects the delicate flower in the bud stage; prepares food via photosynthesis.
2. Petal (Corolla)	Brightly colored to attract insects for pollination; often emits sweet fragrances.
3. Stamen (Androecium)	Male organ that produces pollen grains inside anthers to facilitate fertilization.
4. Pistil (Gynoecium)	Female organ that receives pollen on stigma and houses the ovules where fertilization occurs.

TOPPER DIAGRAM SHEET GUIDE: When drawing a flower diagram, ALWAYS use distinct colors for the Male organs (yellow for anther/filament) and Female organs (green for stigma/style/ovary). Use double-sided arrows to group Stamen (filament + anther) and Pistil (stigma + style + ovary) to show the whorl groupings clearly to the examiner.

ICSE GRADE 5 SCIENCE NOTES

Chapter 5: Plant Reproduction

I. IMPORTANT THINGS TO KNOW (KEY POINTS)

- **Fertilisation** is the fusion of a male gamete with a female gamete inside the ovule to form a single-celled **zygote**.
- Following fertilisation, the **ovule** matures into the **seed**, and the **ovary** swells and ripens into the **fruit**.
- A seed consists of a protective **seed coat** (which has a tiny opening to absorb water) and **cotyledons** (seed leaves) that store food for the baby plant (**embryo**).
- **Germination** is the process of a seed developing into a seedling. It strictly requires three environmental factors: **air (oxygen)**, **water**, and **warmth (warm temperature)**.
- **Seed Dispersal** is the scattering of seeds away from the parent plant. It is essential to prevent overcrowding and competition for sunlight, water, and soil nutrients. Dispersal agents are: **wind, water, animals, and explosion**.
- **Vegetative Reproduction** is the propagation of new plants from vegetative structures: stems (potato, rose, ginger), roots (carrot, sweet potato), or leaves (Bryophyllum). Spores are used by non-flowering plants (mosses, ferns).

II. NEW WORDS DEFINITIONS

- **Seed Coat**: The tough outer protective layer of a seed that protects the internal embryo from injury and drying out.
- **Cotyledons**: The first leaves (seed leaves) of a seed that store food to nourish the growing seedling.
- **Embryo**: The delicate baby plant nestled within the seed, comprising the plumule (future shoot) and radicle (future root).
- **Seedling**: A very young, green plant that has just emerged from the seed coat with its first true leaves.
- **Stratification**: A physiological process where certain seeds must experience cold temperatures before they can break dormancy and germinate.

III. IN-TEXT MIDDLE OF CHAPTER QUESTIONS

Page 66 (T/F Questions):

- **Question**: After fertilization, the seed becomes the ovule. — **Topper Answer: False** (The ovule becomes the seed; the ovary becomes the fruit)
- **Question**: The pollen grains sends out a tiny thread-like tube called a pollen tube. — **Topper Answer: True**

- **Question:** The male and female gamete fuse together to form the cotyledon. — **Topper Answer: False** (They fuse together to form a zygote)
- **Question:** The seed needs air, water, light, and warmth to germinate. — **Topper Answer: True**

Page 67 — List how the seeds of these plants are dispersed:

- **1. Pea seeds: Explosion of the dry fruit pod**
- **2. Dandelion seeds: Wind** (hair-like structures act like a parachute)
- **3. Coconut fruit: Water** (buoyant, light, and fibrous husk)
- **4. Maple seeds: Wind** (wing-like structures aid flight)
- **5. Balsam seeds: Explosion of fruit pod**
- **6. Cocklebur seeds: Animals** (have hooks/spines that stick to fur/skin)

Page 70 — Fill in the blanks:

- Any piece of potato with an **eye bud** can grow into a new plant.
- The **stem** of Hibiscus plant can be used to grow new plant.
- The leaves of Bryophyllum form several buds along its **margins** that can grow into a new plant.
- Mosses reproduce through **spores**.

IV. EXERCISES: SECTION I SOLUTIONS

A. Choose the correct option:

- **Question:** The pollen grains are released from
a. cotyledon b. anther c. filament d. style

Topper Answer: b. anther

- **Question:** Pollen grain germinates and form
a. style tube b. seed coats c. pollen tube d. cotyledons

Topper Answer: c. pollen tube

- **Question:** Which term describes the fertilized cell that will eventually develop into a new plant?
a. Zygote b. Ovule c. Pollen d. Female gamete

Topper Answer: a. Zygote

- **Question:** After germination, a young plant emerges from the seed. This young plant with its first true leaves is known as a/an
a. seedling b. ovule c. cotyledon d. pollen

Topper Answer: a. seedling

- **Question:** Dandelion seeds possess a parachute-like structure that facilitates dispersal by
a. water b. wind c. animals d. explosion

Topper Answer: b. wind

B. Assertion and Reasoning Questions:

• **Question: Assertion (A):** Fertilization is the process in which pollen grains land on the stigma of a flower.

Reason (R): Fertilization leads to the formation of a zygote, which is the first stage of a new plant.

Topper Answer: d. A is False and R is True.

Topper Explanation: The process of pollen grains landing on a stigma is called **pollination**. Fertilization is the distinct internal process of gametic fusion to form a zygote (R is True). Therefore, Assertion A is scientifically false.

• **Question: Assertion (A):** Animals can help disperse seeds.

Reason (R): Some seeds have hooks or spines that can attach to animals.

Topper Answer: a. Both A and R are True and R is the correct explanation of A.

Topper Explanation: Seeds like cocklebur have specialized hook/spine structures. When animals brush past, these hooks cling to their hair/fur, enabling transport and dispersal away from the parent plant.

C. Choose the correct option to fill in the blank:

- A seed has an outer covering called the **seed coat** (seed cover/seed coat).
- The pollen grain may germinate because of the presence of a **sticky** (dry/sticky) substance released by the stigma.
- The zygote develops into a/an **embryo** (embryo/cotyledon) after fertilization.
- Dandelion seeds have hair-like structure that acts like a **parachute** (wing/parachute).
- Dispersal by explosion takes place in **balsam** (balsam/maple) fruit.

D. Give two examples of the following:

- **1. Seeds dispersed by water:** Coconut, Lotus
- **2. Seeds dispersed by animals:** Cocklebur, Xanthium (Tiger claw), Mango (fruit eaten and seed discarded)
- **3. Plants that can grow from stems:** Rose, Potato, Sugarcane, Hibiscus
- **4. Plants that can grow from roots:** Sweet potato, Carrot, Turnip, Dahlia
- **5. Plants that reproduce through spores:** Ferns, Mosses

V. EXERCISES: SECTION II SOLUTIONS

E. Short Answer Questions (Page 71-72):

Q.1) Define fertilization in case of plants.

Answer: Fertilisation is the biological fusion of the male gamete (carried down style by the pollen tube) with the female gamete (egg cell) inside the ovule, resulting in the formation of a single-celled **zygote**.

Q.2) What is a seed coat? What is the role of the tiny hole that is present on the seed coat?

Answer: The seed coat is the tough outer protective covering of a seed. The tiny hole (called the **micropyle**) allows water and oxygen to enter the seed coat, triggering metabolic activity necessary for germination.

Q.3) When a seed has two cotyledons, why are they thick and fleshy?

Answer: They are thick and fleshy because they store food reserves in the form of starch and nutrients. This stored food nourishes the growing baby plant (embryo) until it develops roots and green leaves to make its own food via photosynthesis.

Q.4) What do you understand by germination of a seed?

Answer: Germination is the physiological process by which an inactive, dormant seed activates and grows into a young seedling when placed in favorable conditions of water, air, and warmth.

Q.5) A farmer's corn seeds struggle to sprout despite good weather. Why do all seeds not grow into new plants?

Answer: Not all seeds germinate because: 1) Many seeds are consumed by birds and insects. 2) Heavy rain, strong wind, or freezing frost destroy them. 3) Some seeds lack viable embryos or get infected by fungal diseases.

Q.6) Which part of a rose plant can be used to grow new plants?

Answer: A **stem cutting** containing at least one healthy node/lateral bud can be used to grow a new rose plant through vegetative reproduction.

F. Long Answer Questions (Page 72):

Q.1) How does formation of fruits and seeds take place after fertilization?

Answer: Once fertilisation takes place, the flower undergoes rapid anatomical transformations:

- 1) The single-celled **zygote** divides repeatedly to form the **embryo** (baby plant).
- 2) The outer layers of the **ovule** harden and dry up to form the protective **seed coat**. The entire ovule becomes the **seed**.
- 3) The **ovary** absorbs nutrients, swells, ripens, and transforms into the **fruit** (which protects the seeds inside).
- 4) Other floral structures, including the petals, sepals, stigma, and style, wither, dry, and fall away from the base.

Q.2) Describe the structure of a bean seed.

Answer: A bean seed is a typical dicotyledonous seed consisting of the following key parts:

- 1) **Seed Coat (Testa):** The dark, tough outer protective covering.
- 2) **Micropyle (Tiny Hole):** A minute opening near the scar (hilum) which absorbs water to swell the seed.
- 3) **Two Cotyledons:** Large, fleshy seed leaves that occupy most of the seed volume and store protein and starch.
- 4) **Embryo:** The delicate baby plant positioned between the cotyledons. It consists of two parts:
 - **Radicle:** The lower embryonic tip which grows downwards to form the primary root system.
 - **Plumule:** The upper embryonic tip which grows upwards to form the primary shoot system.

Q.3) Why is there a need for seed dispersal?

Answer: If all seeds produced by a plant were to fall directly beneath it, they would germinate in a tight cluster. This would lead to severe **overcrowding** and intense competition for limited resources like sunlight, water, space, and mineral nutrients. Under such competition, most seedlings would wither and die. Dispersal distributes seeds to new geographical areas, ensuring survival, expansion, and minimal parent-offspring competition.

Q.4) Explain with examples how seed dispersal by wind takes place.

Answer: Seeds dispersed by wind have structural adaptations that make them light and aerodynamically suited to drift in the air:

- 1) **Hair-like Structures:** Dandelion seeds possess fine, silky, feather-like hairs that form a parachute. This catches breeze currents, allowing the seed to float long distances. (Example: Dandelion).
- 2) **Wing-like Structures:** Some seeds have flat, papery, wing-like expansions. When the fruit splits, these wings spin in the wind like helicopter blades, carrying the seed far away. (Example: Maple, Drumstick, Jacaranda).

Q.5) Describe other ways of reproduction in plants apart from seeds.

Answer: Many plants reproduce without seeds through **vegetative propagation** and **spore formation**:

- 1) **Stems:** Underground stems have buds ('eyes') that grow into new plants (Potato, Ginger). Runners and stem cuttings with buds planted in soil grow roots and shoots (Rose, Hibiscus, Sugarcane).
- 2) **Roots:** Tuberous roots store starch and develop shoots when placed in water/damp soil (Sweet Potato, Carrot, Dahlia).
- 3) **Leaves:** Leaf margins contain tiny vegetative buds. When a leaf falls on moist soil, these buds grow roots, detach, and form independent plantlets (Bryophyllum).
- 4) **Spores:** Non-flowering plants do not produce seeds. Instead, they produce minute, microscopic dust-like reproductive bodies called **spores** inside specialized capsules. (Examples: Mosses, Ferns).

VI. PICTURE STUDY & WORKSHEET SOLUTIONS (Page 72-75)

Picture Study: Fertilisation in Plants

1. What does the given picture show?

Answer: The picture shows the process of **Fertilisation in a flowering plant**, specifically depicting a pollen tube carrying male gametes to fertilize the egg cell inside the ovule.

2. Label the parts (i), (ii), (iii), and (iv) (Page 73):

- (i) Ovule
- (ii) Female egg cell / gamete (inside the ovule)
- (iii) Pollen tube (growing down the style)
- (iv) Pollen grain (germinating on the stigma)

3. On germination, what comes out from the pollen grain?

Answer: A long, thin, thread-like tube called the **pollen tube** comes out.

4. Where does it pass through? What does it contain?

Answer: It passes through the long, hollow style of the gynoecium to reach the ovary. It contains **male gametes (reproductive cells)**.

5. What is formed after the fusion of male and female gamete? Where does it occur and what is this process called?

Answer: A **zygote** is formed. This fusion occurs inside the **ovule**, and the process is called **fertilisation**.

Worksheet: Complete the Sentences (Page 75)

Section 1: Coconut

1. This is a **germinating coconut palm** seed.
2. Its fruits have a **fibrous** covering.
3. It helps it to **float easily on water**.
4. Dispersal of coconut thus takes place by **water**.

Section 2: Cocklebur

1. This is a **cocklebur** plant.
2. The seeds of cocklebur plant have **hooks or spines**.
3. These seeds thus stick to the **bodies or fur of passing animals**.
4. So the dispersal of seeds takes place by **animals**.

Section 3: Sweet Potato

1. This is a figure of a **sweet potato**.
2. It shows vegetative propagation through **roots**.
3. Other plants that also show vegetative propagation through roots are:
 - (a) **Carrot**
 - (b) **Turnip**
 - (c) **Dahlia / Radish**

TOPPER EXAM DRAWING TIP (GERMINATION): Always draw the 'Three-Beak Seed Germination Experiment' diagram if asked about conditions of germination. Show Seed 1 (top, only air), Seed 2 (middle, gets both water and air - germinated), and Seed 3 (bottom, submerged in water, only water, no air). This setup proves that ALL THREE factors (air, water, and warmth) are collectively required.

ICSE GRADE 5 SCIENCE NOTES

Chapter 6: Solids, Liquids, and Gases

I. IMPORTANT THINGS TO KNOW (KEY POINTS)

- **Matter** is defined as anything that has mass and occupies physical space. It exists in three states: solid, liquid, and gas.
- **Solids** have a fixed shape and volume because their constituent molecules are tightly packed with virtually no intermolecular space and very high force of attraction.
- **Liquids** have a fixed volume but no fixed shape, taking the shape of their container. Their molecules are loosely packed with moderate intermolecular spaces.
- **Gases** have neither a fixed shape nor a fixed volume, expanding to completely fill any available space. Their molecules are very loosely packed with extremely weak forces.
- **Solutions** are formed when a solute (dissolving substance) dissolves in a solvent (the dissolving medium). Water is known as the **universal solvent**.
- **Separation of Soluble Substances** is achieved by heating: **boiling** (leaving solute behind) or **distillation** (collecting both pure solute and pure solvent by evaporation and condensation).
- **Separation of Insoluble Substances** utilizes size or density: **sedimentation & decantation** (settling and pouring) or **filtration** (passing through filter paper).
- **Air** is a mixture of gases (78% Nitrogen, 21% Oxygen, 1% carbon dioxide, argon, water vapor). Moving air is called **wind**.
- Heating causes air to expand and become lighter, making warm air rise. This temperature-driven density change creates **sea breezes** (daytime sea-to-land) and **land breezes** (nighttime land-to-sea).

II. NEW WORDS DEFINITIONS

- **Molecules**: The tiny particles that make up all matter.
- **Solute**: The solid or gas substance that dissolves in a liquid to form a solution (e.g. salt).
- **Solvent**: The liquid medium in which a solute dissolves (e.g. water).
- **Miscible Liquids**: Liquids that mix completely and seamlessly with each other (e.g. alcohol and water).
- **Immiscible Liquids**: Liquids that do not mix and form separate layers (e.g. oil and water).
- **Humidity**: The total amount of moisture or water vapor present in the atmosphere at a specific time.
- **Filtration**: The mechanical separation of insoluble solids from a liquid by passing it through porous filter paper.

III. IN-TEXT MIDDLE OF CHAPTER QUESTIONS

Page 79 (T/F Questions):

- **Question:** Copper sulphate crystals are an example of solids. — **Answer: True**
- **Question:** In solid state of matter, the molecules are tightly packed. — **Answer: True**
- **Question:** Petrol is a liquid state as it does not have a fixed shape, but has a fixed volume. — **Answer: True**
- **Question:** The liquid state of matter can fill the entire space available to it. — **Answer: False** (Only gases expand to fill all available space)
- **Question:** Oxygen and nitrogen are solid states of matter. — **Answer: False** (They are gaseous states)

Page 83 — Fill in the blanks:

- The substance that dissolves completely in a liquid, to form a solution is called **solute**.
- Tea leaves and sand are **insoluble** substances in water.
- Water is lost through **boiling**.
- Distilled water is not suitable for **drinking**.
- The process of separating insoluble substances by passing the solution through filter paper is called **filtration**.

Page 87 (T/F Questions):

- **Question:** Air is a mixture of several gases such as nitrogen, oxygen, carbon dioxide, etc. — **Answer: True**
- **Question:** Air contains dust particles and smoke. — **Answer: True**
- **Question:** Oxygen is taken in by green plants to prepare food. — **Answer: False** (Plants absorb carbon dioxide for photosynthesis; oxygen is released)
- **Question:** Nitrogen in air controls the process of burning. — **Answer: True** (Oxygen supports burning; nitrogen acts as a buffer to slow it down)
- **Question:** Air has no weight. — **Answer: False** (Air is matter and possesses weight)
- **Question:** A gentle wind is called gale. — **Answer: False** (A gentle wind is called a breeze; a gale is a very strong, violent wind)
- **Question:** The land breeze takes place during daytime. — **Answer: False** (Land breeze takes place during the night)
- **Question:** Summer monsoon wind brings rain in India. — **Answer: True**

IV. EXERCISES: SECTION I SOLUTIONS

A. Choose the correct option to fill in the blank (Page 90):

- Matter is made of molecules that are very **small** (large/small).
- In solids, there is **no** (no/some) space between the molecules.
- Copper sulphate crystals are **solids** (solids/liquids).

- Solids and **liquids** (gases/liquids) have fixed volume.
- Solute and **solvent** (gas/solvent) makes a solution.

B. Assertion and Reasoning Questions (Page 91):

- **Question: Assertion (A):** Filtration can be used to separate a mixture of sand and water.

Reason (R): Sand particles are smaller than water molecules and can pass through the filter paper.

Topper Answer: c. A is True and R is False.

Topper Explanation: Assertion A is completely true. Reason R is false because filtration works precisely because sand particles are much **larger** than water molecules and filter paper pores, allowing them to be trapped on top while water flows through.

- **Question: Assertion (A):** Ventilators help maintain good air quality in closed spaces.

Reason (R): They only remove warm air and replace it with cooler air.

Topper Answer: a. Both A and R are True and R is the correct explanation of A.

Topper Explanation: Warm air, being lighter, naturally rises up. Ventilators located near ceilings allow this warm, stale air to exit, which is continually replaced by cooler, fresh oxygen-rich air coming through lower doors and windows.

C. Match the following (Page 91):

1. Gas — **iii. Flows in all direction**
2. Solution — **v. Sugar or salt dissolved in water**
3. Chalk — **iv. Insoluble in water**
4. Pure water — **ii. Obtained by distillation process**
5. Honey — **i. Soluble in water**

V. EXERCISES: SECTION II SOLUTIONS

D. Short Answer Questions (Page 91):

Q.1) Is the arrangement of molecules in all states of matter the same?

Answer: No. In **solids**, molecules are tightly packed with almost no space between them. In **liquids**, they are loosely packed with small spaces. In **gases**, they are very loosely packed with massive spaces between them.

Q.2) What is a solution? Give two examples.

Answer: A solution is a uniform, homogeneous mixture formed when a soluble substance (solute) dissolves completely in a liquid (solvent). Examples: Sugar solution, Salt solution, Carbonated soda water.

Q.3) Shubham is making hot chocolate. Cocoa powder mixes smoothly with hot milk, but marshmallows float on top. What kind of substances are they?

Answer: Cocoa powder is a **soluble substance** because it dissolves completely in the warm milk solvent. Marshmallows are **insoluble substances** that do not dissolve and float because they are less dense than the milk.

Q.4) Neha has rainwater with dirt and leaves. How can she remove them so she can clean the terrace floor? What is this process called?

Answer: She can remove the dirt and leaves by pouring the mixture through a fine mesh strainer or filter paper. This separation process is called **filtration** (or sedimentation and decantation).

Q.5) What are the different properties of air?

Answer: Air possesses four key physical properties:

- 1) Air occupies physical space.
- 2) Air exerts pressure in all directions.
- 3) Air has weight (mass).
- 4) Air is required for combustion (burning).

E. Long Answer Questions (Page 92):

Q.1) Write three properties each of solids, liquids, and gases. Can you think of everyday objects that demonstrate these properties?

Answer: Solids: 1) Fixed shape. 2) Fixed volume. 3) Do not flow (molecules tightly packed). Objects: A wooden table, a school textbook.

Liquids: 1) No fixed shape (take shape of container). 2) Fixed volume. 3) Flow from high to low levels. Objects: Drinking water, milk, cooking oil.

Gases: 1) No fixed shape. 2) No fixed volume (fill entire container space). 3) Flow freely in all directions. Objects: Air inside a bicycle tire, gas inside an LPG cooking cylinder.

Q.2) Write an activity to identify soluble and insoluble substances in water.

Answer: Objective: To classify substances as soluble or insoluble in water.

Materials: Water, clear glasses, sugar, sand, salt, chalk powder, stirrer.

Procedure: Fill four glasses half with water. Add a spoonful of sugar in Glass 1, sand in Glass 2, salt in Glass 3, and chalk powder in Glass 4. Stir each glass with the stirrer and leave undisturbed for 5 minutes.

Observation: Sugar and salt completely disappear in Glass 1 and 3, forming clear liquids. Sand and chalk powder settle down at the bottom in Glass 2 and 4, and the water remains turbid.

Conclusion: Substances that dissolve completely (sugar, salt) are **soluble**. Substances that do not dissolve and settle down (sand, chalk) are **insoluble**.

Q.3) Explain two methods to separate soluble substances.

Answer: 1) Boiling (Evaporation): The solution is heated. The liquid solvent (water) evaporates into the air as vapor, leaving behind the solid solute (salt/sugar) as dry crystals at the bottom.

2) Distillation: The solution is boiled in a round-bottomed flask. The solvent evaporates into vapor, which is channeled through a cooling condenser. The condenser cools the vapor back into pure liquid water, which is collected in a separate flask. The pure solute remains in the boiling flask. This allows us to recover both solute and solvent.

Q.4) How are sea breeze and land breeze formed?

Answer: Breezes are formed due to unequal heating and cooling rates of land and water:

• **Sea Breeze (Daytime):** During the day, land heats up much faster than the sea. The air over the land becomes warm, expands, and rises. Cooler air from over the sea rushes toward the land to fill the vacant

space. This cooler daytime sea-to-land wind is a sea breeze.

• **Land Breeze (Nighttime):** At night, land loses heat much faster than the sea. Therefore, the air over the sea remains warmer and rises. Cooler air from over the land blows toward the sea to replace it. This cooler nighttime land-to-sea wind is a land breeze.

Q.5) What are monsoon winds? How are they helpful to us?

Answer: Monsoon winds are heavy, seasonal, moisture-laden winds. In summer, the land masses of India heat up intensely, creating low pressure. This causes cool, moisture-packed winds to blow from the Indian Ocean toward the hot mainland, bringing widespread rainfall. They are incredibly helpful because they provide vital rainwater for agriculture (watering crops like rice) and lower the intense summer temperatures.

VI. PICTURE STUDY & WORKSHEET SOLUTIONS (Page 92-94)

Picture Study: Sea and Land Breeze Analysis

1. Does the picture (depicting a hot day with arrows moving from the ocean to the house on land) depict sea breeze or land breeze?

Answer: The picture depicts a **Sea Breeze**.

2. When does it occur—day or night? Draw a sun or moon accordingly.

Answer: It occurs during the **daytime**. A bright sun should be drawn above the scene.

3. Explain briefly what is happening in the picture:

Answer: During the hot daytime, solar heat warms the land faster than the seawater. The air immediately above the land heated by conduction expands and rises. This creates a local drop in air pressure on land, which draws the cooler, high-pressure air from the sea onto the land. This creates a refreshing sea breeze.

Worksheet: Molecular Arrangements Table (Page 94)

Molecular Drawing Pattern	Identify State	Three Core Examples	Three Key Characteristics
Loosely packed, scattered circles with gaps	Liquid	Water, milk, petrol, oil	1) No fixed shape (fluid). 2) Fixed volume. 3) Molecules flow easily from high to low level.
Very loosely packed, far apart spheres with arrows	Gas	Air, oxygen, steam, carbon dioxide	1) No fixed shape. 2) No fixed volume (fills container). 3) Extremely high flow in all directions.
Tightly packed spheres arranged in a rigid cubic block	Solid	Ice, stone, copper sulphate, wood	1) Fixed, rigid shape. 2) Fixed volume. 3) Molecules do not flow (only vibrate).

TOPPER CORE DRAWING GUIDE: To draw molecular arrangement circles: Use a compass or stencil. For solids, circles **MUST** touch each other in rows (lattice). For liquids, draw them close but slightly overlapping/irregular with small spaces. For gases, draw only 5-6 circles spaced very far apart with motion lines. Do not draw them randomly!

TREE HOUSE HIGH SCHOOL • MID-TERM REVIEW KEY

Mid-Term Revision Worksheet Solutions

SECTION A — SHORT ANSWER QUESTIONS (1–2 Sentences)

Q.1) What are nutrients? Name any four nutrients present in food.

Answer: Nutrients are essential chemical compounds present in food that provide the body with energy, support growth, and assist in disease prevention. Four main nutrients are **carbohydrates, proteins, fats, and vitamins**.

Q.2) What is a balanced diet? Why is it important for our health?

Answer: A balanced diet is a diet containing all components of food in the right proportions for the healthy functioning of the body. It is important because it promotes proper development of body and mind, prevents deficiencies, and maintains body weight.

Q.3) What are deficiency diseases? Give one example.

Answer: Deficiency diseases are non-communicable diseases caused by a persistent lack of a specific nutrient (vitamin or mineral) in a person's diet over a long period. Example: **Scurvy** (caused by lack of Vitamin C).

Q.4) What is pollination? Name the two types of pollination.

Answer: Pollination is the transfer of pollen grains from the male anther to the female stigma of a flower. The two types are **self-pollination** and **cross-pollination**.

Q.5) Differentiate between self-pollination and cross-pollination.

Answer: Self-pollination is the transfer of pollen within the same flower or flowers on the same plant. Cross-pollination is the transfer of pollen from the anther of one flower to the stigma of a different flower on a different plant of the same kind.

Q.6) What is fertilisation in plants?

Answer: Fertilisation in plants is the fusion of the male gamete (pollen nucleus) with the female gamete (egg cell) inside the ovule of a flower to form a single-celled **zygote**.

Q.7) What is the function of a seed in a plant?

Answer: The primary function of a seed is to protect and nourish the plant embryo (baby plant) inside it and to germinate into a new plant under favorable environmental conditions, ensuring species continuity.

Q.8) Name the three states of matter.

Answer: The three states of matter are **solids, liquids, and gases**.

Q.9) What is the difference between evaporation and condensation?

Answer: Evaporation is the physical process where a liquid changes into a gas upon heating (e.g., water turning to steam). Condensation is the process where a gas cools down and converts back into a liquid (e.g., steam turning into water).

Q.10) Why does a liquid take the shape of the container in which it is kept?

Answer: A liquid takes the shape of its container because its constituent molecules are loosely packed with weak forces of attraction, allowing them to flow and slide over one another easily.

SECTION B — LONG ANSWER QUESTIONS (4–6 Sentences)

Q.1) Explain the different nutrients in food and describe the importance of carbohydrates, proteins, fats, vitamins and minerals.

Answer: Food contains five primary groups of nutrients necessary for life:

- 1) **Carbohydrates:** These are the body's primary fuel source, providing instant energy to carry out daily tasks. Sources: rice, potatoes.
- 2) **Fats:** They serve as an energy reserve, offering more energy than carbs, and form an insulating layer under the skin to keep the body warm. Sources: butter, nuts.
- 3) **Proteins:** They are called muscle-builders, vital for growth and cellular repair of injured tissues. Sources: eggs, pulses.
- 4) **Vitamins & Minerals:** Known as protective foods. Needed in micro-quantities to strengthen immunity, fight infections, maintain bone/tooth structure, and assist blood formation. Sources: fruits, leafy green vegetables.

Q.2) What is a balanced diet? Explain why a balanced diet, regular exercise, cleanliness and adequate sleep are important for maintaining good health.

Answer: A balanced diet provides all necessary nutrients in exact proportions. Maintaining good health requires a combination of healthy habits:

- **Balanced Diet:** Ensures our organs have proper nutrition for metabolic processes without gaining unhealthy fat.
- **Regular Exercise:** Keeps muscles toned, builds strong bone mass, improves blood circulation, and releases stress.
- **Cleanliness:** Prevents the breeding and transmission of disease-causing pathogens and household germs, reducing infections.
- **Adequate Sleep:** Allows the brain to process memory and gives cells time to rest, regenerate tissues, and restore vital energy.

Q.3) What are deficiency diseases? Explain any three deficiency diseases, along with the nutrients whose deficiency causes them.

Answer: Deficiency diseases occur when a vital nutrient is severely lacking in our food over a long duration. Three classic examples are:

- 1) **Scurvy (Vitamin C Deficiency):** Leads to weak, bleeding gums, spongy teeth, and incredibly slow wound healing. Prevented by eating citrus fruits like amla and lemon.
- 2) **Rickets (Vitamin D Deficiency):** Causes bone softening, bowlegs, and knock-knees. Prevented by morning sun exposure and eating eggs/fish.
- 3) **Goitre (Iodine Deficiency):** Causes severe swelling of the thyroid gland in the neck and retards

mental/physical growth. Prevented by cooking with iodized table salt.

Q.4) Explain the process of pollination. Describe how insects, birds, wind and other agents can help in carrying pollen from one flower to another.

Answer: Pollination is the initial physical step of plant reproduction. Moving non-motile pollen grains from anthers to stigmas requires vector agents:

- **Insects and Birds:** Scented flowers with sweet nectar attract bees, moths, and hummingbirds. While they feed, heavy sticky pollen grains cling to their hairy limbs and bodies. When they fly to another flower, this pollen rubs off onto the sticky stigma.
- **Wind:** Some plants (grasses) have lightweight, dry, powdery pollen. Wind currents blow these grains away. Receptive flowers have feathery stigmas that act like nets to catch the drifting pollen.
- **Water:** Aquatic plants shed pollen directly into water currents, which carry them downstream to float into contact with other female flowers.

Q.5) Explain the difference between self-pollination and cross-pollination. Give one advantage of cross-pollination.

Answer: Self-pollination involves pollen transfer within the same flower or within flowers on a single parent plant, requiring no external agents. In contrast, cross-pollination involves transferring pollen grains from one plant's anther to a different plant's stigma of the same species, requiring insects or wind vectors.

Advantage of Cross-Pollination: It produces offspring with greater genetic variation, resulting in healthier, sturdier plants that possess stronger disease resistance and better environmental adaptability.

Q.6) Describe the process of reproduction in flowering plants, beginning with pollination and ending with the formation of seeds and fruits.

Answer: Reproduction in flowering plants proceeds in a highly coordinated sequence:

- 1) **Pollination:** Pollen grains are transferred from the male anther onto the sticky female stigma.
- 2) **Pollen Tube Growth:** The pollen grain germinates on the wet stigma and grows a microscopic **pollen tube** down style to the ovary.
- 3) **Fertilisation:** Male gametes travel down the tube and fuse with the female egg cell inside the ovule, forming a **zygote**.
- 4) **Maturation:** The zygote grows into an **embryo** inside the ovule. The ovule develops a tough seed coat and becomes the **seed**.
- 5) **Fruit Formation:** The ovary grows rapidly, ripens, and turns into the **fruit**, while petals and sepals fall off.

Q.7) What are the different methods by which seeds are dispersed? Explain any four methods with suitable examples.

Answer: Dispersal is critical to prevent seedling overcrowding. The four primary methods are:

- 1) **Dispersal by Wind:** Lightweight seeds with hair or wings fly in the breeze. E.g., **Dandelion seeds** (silky hairs), **Maple seeds** (wings).
- 2) **Dispersal by Water:** Aquatic or riverbank plants have light, buoyant seeds that float downstream. E.g., **Coconut** (fibrous husk), **Lotus** (spongy float).
- 3) **Dispersal by Animals:** Seeds have hooks to cling to animal fur, or are enclosed in delicious fruits that animals consume, passing the hard seeds undigested in waste. E.g., **Cocklebur** (hooks), **Mango / Guava** (eaten by birds).
- 4) **Dispersal by Explosion:** Some fruit pods dry, build tension, and split open violently, throwing seeds outwards. E.g., **Pea**, **Balsam**.

Q.8) Explain the characteristics of solids, liquids and gases. How are the particles in these three states different?

Answer: The properties of the three states of matter are determined by their molecular packing:

- **Solids:** Have a definite shape and a definite volume. Their molecules are highly packed in a tight, orderly grid with massive attraction forces, allowing them to only vibrate in place.
- **Liquids:** Have no fixed shape (take container shape) but possess a fixed volume. Their molecules are loosely packed with moderate spaces, allowing them to flow and slide over one another easily.
- **Gases:** Have neither a fixed shape nor volume. Their molecules are extremely far apart with virtually non-existent attraction, allowing them to move freely at high speeds in all directions.

Q.9) Describe what happens when a substance changes from one state of matter to another. Explain melting, freezing, evaporation and condensation with examples.

Answer: Phase changes are physical processes driven by the addition or removal of thermal energy:

- **Melting (Solid to Liquid):** Heating a solid adds energy, causing molecules to break their rigid grid. Example: Ice cube melting into liquid water when removed from the freezer.
- **Freezing (Liquid to Solid):** Cooling a liquid removes energy, causing molecules to slow down and bond rigidly. Example: Water turning into hard ice in a freezer.
- **Evaporation (Liquid to Gas):** Heating a liquid drives molecules far apart to fly as gas. Example: Water boiling to form steam.
- **Condensation (Gas to Liquid):** Cooling a gas slows molecules down, bringing them close together. Example: Water vapor condensing into dew drops on cold grass.

Q.10) Explain how heating and cooling can cause changes in the state of matter. Give examples from everyday life.

Answer: State changes are direct results of molecular energy changes caused by heating and cooling:

- **Heating (Expands and loosens):** Heating increases kinetic energy of molecules, breaking their tight packing. Solid becomes liquid (melting), and liquid becomes gas (evaporation). Everyday examples: Melting of solid butter on a hot pan, boiling water in a tea kettle.
- **Cooling (Contracts and aggregates):** Cooling decreases kinetic energy, causing molecules to contract, lose motion, and bind tightly together. Gas turns to liquid (condensation), and liquid turns to solid (freezing). Everyday examples: Frost forming on refrigerator coils, water droplets forming on the outer surface of a glass filled with ice cubes.

THINK AND ANSWER ■ (ANALYTICAL QUESTIONS)

Question: A glass contains some ice cubes. After some time, the ice cubes disappear and water is left in the glass. Later, the water is left in the sun and slowly disappears.

- a) What change happens when ice becomes water?
- b) What change happens when water becomes water vapour?
- c) Is either of these changes reversible? Explain your answer.

Topper Solution & Scientific Explanation:

- **(a) Melting (Solid to Liquid Phase Transition):** Heat from the warm surrounding air enters the glass. This thermal energy increases the kinetic motion of the tightly packed water molecules in ice, allowing them to break their rigid crystalline lattice structure and flow freely as liquid water.

- **(b) Evaporation (Liquid to Gas Phase Transition):** When placed under the sun, water absorbs radiant solar energy. The surface molecules gain enough kinetic energy to break their liquid-state intermolecular forces and escape into the atmosphere as invisible water vapor.

- **(c) Yes, BOTH of these physical changes are completely REVERSIBLE.**

These are physical phase changes because the chemical identity of the molecules remains identical (H_2O) across all three states—solid ice, liquid water, and gaseous steam. We can easily reverse the evaporation process through **condensation** (cooling water vapor back into liquid water) and reverse the melting process through **freezing** (cooling liquid water back into solid ice cubes).